

HEALTHCONNECT CLINIC

PROJECT SUMMARY

Problem

The HealthConnect project focuses on understanding patient appointment attendance and identifying factors associated with missed appointments. The aim is to use data analysis to uncover useful patterns that can support improved appointment management and patient engagement.

Resources Used

The main resources reviewed were the HealthConnect Appointment Dataset, the HealthConnect Data Dictionary, and the HealthConnect Clinic Knowledge Base. These resources were used to understand the available data, variable definitions and the business context of the project.

Key Observations

The appointment dataset contains 5,000 records across 18 variables and 1,696 unique patients. The recorded appointment outcomes include Attended, No-Show and Cancelled.

The initial data-quality review identified missing values in `reminder_channel`, `distance_to_clinic_km` and `waiting_time_minutes`. The missing reminder-channel values correspond with appointments where no reminder was recorded. A date-format issue was also identified in `booking_date`, which will require standardisation and validation before further date-based analysis.

Proposed Approach

The dataset will first be cleaned and validated using Excel and Power Query. Exploratory analysis will then be carried out in Power BI, focusing on appointment outcomes, booking lead time, previous no-shows, reminder activity, appointment characteristics, timing, distance and waiting time.

Five potential KPIs have been identified: No-Show Rate, Appointment Attendance Rate, Repeat No-Show Rate, Reminder-Associated Attendance Rate and Average Booking Lead Time.

Key Considerations

The analysis will take into account missing values, date-format consistency, unequal group sizes and the limitations of the available variables. Any relationships identified will be treated as associations rather than automatically interpreted as causal relationships.

Week 5 Focus

The next stage will focus on deeper exploratory analysis of appointment attendance and no-show patterns, including KPI development, segmentation and visual analysis. The findings from this stage will help determine the most relevant insights for the HealthConnect project.